WORKED SOLUTION

STEP II 1987, Question 16

Probability and Statistics • Bayes' Theorem

STEP II 1987, Question 16

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s2-q16

The key idea

This is Bayes' theorem in its plainest form. Two reports of "yes" can arise in two ways: the claim is true and both friends are telling the truth, or it is false and both are lying. We weigh each route by its prior probability and the probability of that pair of reports, then ask what fraction of the total comes from the "true" route. The second part works the same way. The only new point is that the friends' answers say something about A alone, so we must sort the three possibilities by whether A is in the fridge.

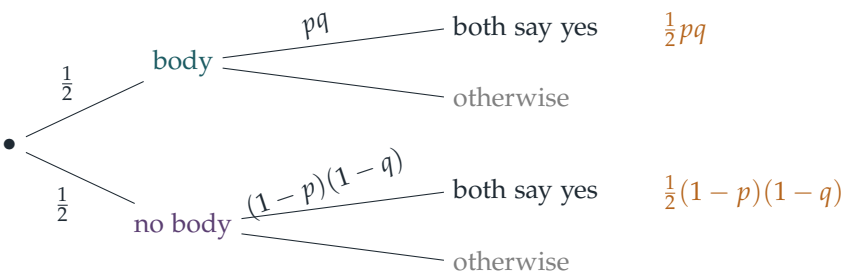
Solution

The first estimate

Let F be the event that there is a body in the fridge, and let Y be the event that both P and Q say there is. This morning $\Pr(F) = \Pr(F') = \frac{1}{2}$.

We model each friend as answering a yes/no question: telling the truth means giving the correct answer and lying means giving the wrong one. The friends' honesty does not depend on what is in the fridge, and P and Q act independently. So:

- if there is a body, both say "yes" exactly when both tell the truth, so $\Pr(Y \mid F) = pq$;
- if there is no body, both say "yes" exactly when both lie, so $\Pr(Y \mid F') = (1-p)(1-q)$.



By the definition of conditional probability (Bayes' theorem),

$$\Pr(F \mid Y) = \frac{\Pr(F)\Pr(Y \mid F)}{\Pr(F)\Pr(Y \mid F) + \Pr(F')\Pr(Y \mid F')} = \frac{\frac{1}{2}pq}{\frac{1}{2}pq + \frac{1}{2}(1-p)(1-q)}$$
$$= \frac{pq}{pq + 1 - p - q + pq} = \frac{pq}{1 - p - q + 2pq},$$

which is the required estimate. The factors of $\frac{1}{2}$ cancel because the prior was even.

Ask yourself Why isn't the answer simply pq, the chance that both are telling the truth?

Because we already know something: they agreed. The outcomes in which exactly one of them lies are now impossible, as those would give one "yes" and one "no". We have to rescale over what is left. Of the two surviving routes, "both truthful" has weight pq and "both lying" has weight $(1-p)(1-q)$. Dividing by their sum is exactly that rescaling. (The $\frac{1}{2}$ priors play a part too. With an uneven prior they would not cancel.)

The second estimate

Now there is certainly a body. The three equally likely possibilities, each with prior probability $\frac{1}{3}$, are

$$A \text{ only}, \qquad B \text{ only}, \qquad \text{both } A \text{ and } B.$$

Let Y_A be the event that both friends say A is in the fridge. Their answers depend only on whether A is there, by the same model as before:

- in the cases "A only" and "both", A is in the fridge, so $\Pr(Y_A) = pq$ in each;
- in the case "B only", A is not there, so $\Pr(Y_A) = (1-p)(1-q)$.

Hence

$$\Pr(\text{both} \mid Y_A) = \frac{\frac{1}{3}pq}{\frac{1}{3}pq + \frac{1}{3}(1-p)(1-q) + \frac{1}{3}pq}$$
$$= \frac{pq}{2pq + 1 - p - q + pq} = \frac{pq}{1 - p - q + 3pq}.$$

Notice that the reports tell us nothing directly about B. Their only effect is to change the chance that A is present. By the first part (with prior $\frac{2}{3}$ in place of $\frac{1}{2}$), that chance becomes $2pq/(1-p-q+3pq)$. Given that A is present, "A only" and "both" remain equally likely, so we halve it. This gives the same answer.

Checks

- If $p = q = \frac{1}{2}$ the friends are useless, and the answers return the priors: $\frac{1}{4}/\frac{1}{2} = \frac{1}{2}$ and $\frac{1}{4}/\frac{3}{4} = \frac{1}{3}$.

- If $p = q = 1$ the friends are infallible, and the answers become 1 (there is a body) and $\frac{1}{2}$ (A is there, so it's a toss-up whether B is too).
- With $p = q = \frac{3}{4}$ the answers are $\frac{9}{10}$ and $\frac{9}{19}$. Exact enumeration over every combination of state and truth-telling agrees for 200 random pairs (p, q). A simulation of 4×10^5 trials with $p = 0.7$, $q = 0.6$ gave 0.4375 for the second part, matching $\frac{0.42}{0.96} = 0.4375$.

Common slips

- Answering pq, and forgetting that "both lying" also produces two "yes" answers.
- In the second part, putting the "both" case alone in the numerator but leaving out the "A only" case from the denominator. Both cases are consistent with A being present.

Beyond STEP

This is Bayesian updating. A prior belief is multiplied by the *likelihood* of the evidence and then normalised. A tidy way to write it is in odds form: posterior odds equal prior odds times the likelihood ratio $\frac{pq}{(1-p)(1-q)}$. That makes it clear that independent witnesses simply multiply their ratios. The same calculation underlies medical testing, spam filters and the combining of evidence in court. Blitzstein and Hwang, *Introduction to Probability*, has an excellent chapter on conditional probability with many puzzles of exactly this flavour.

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